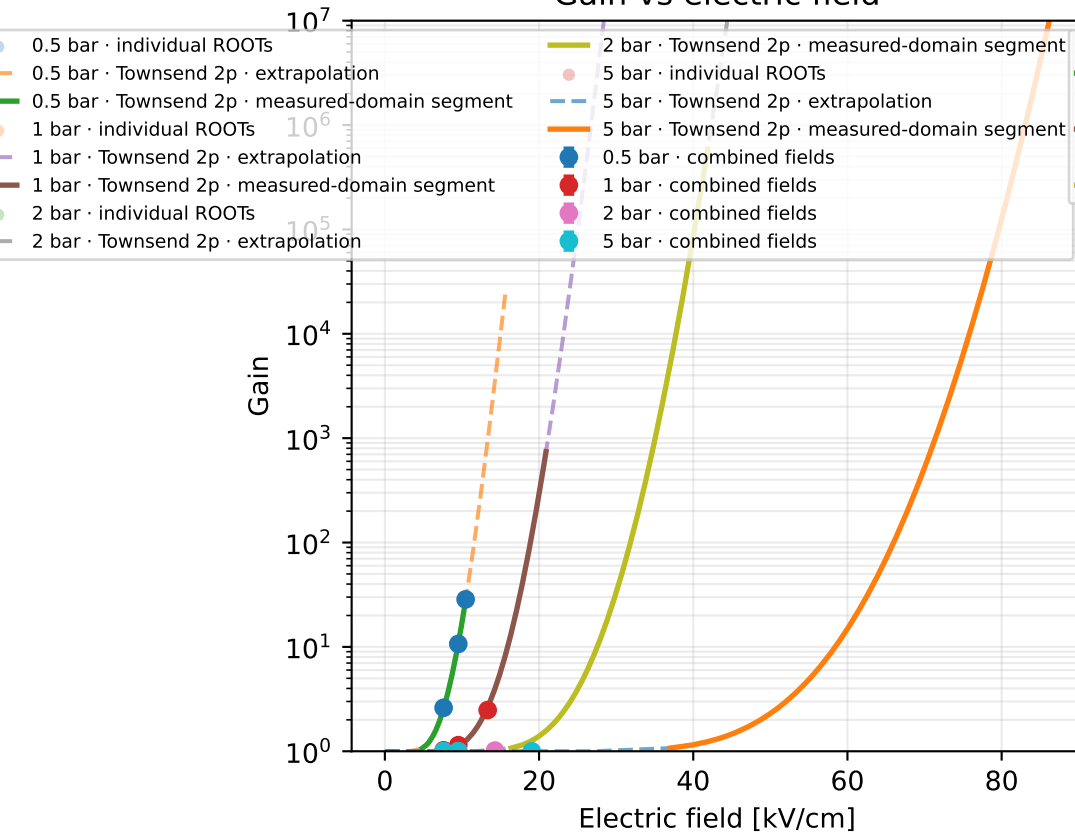
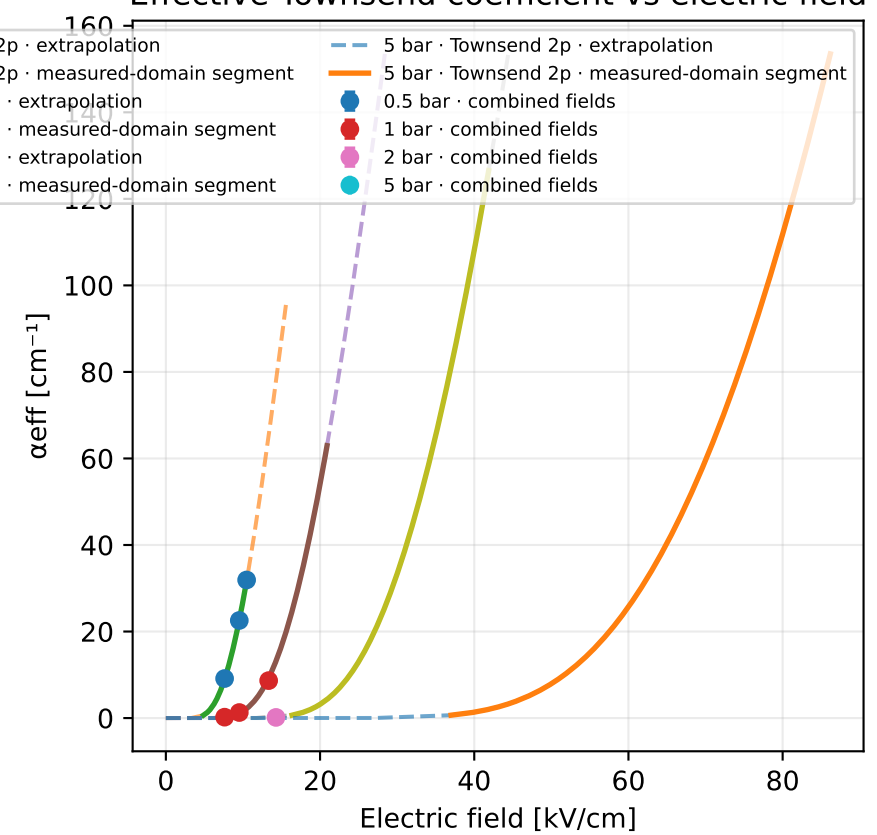


ArN2 · Ar 90 % / N2 10 % · gap 1.05 mm

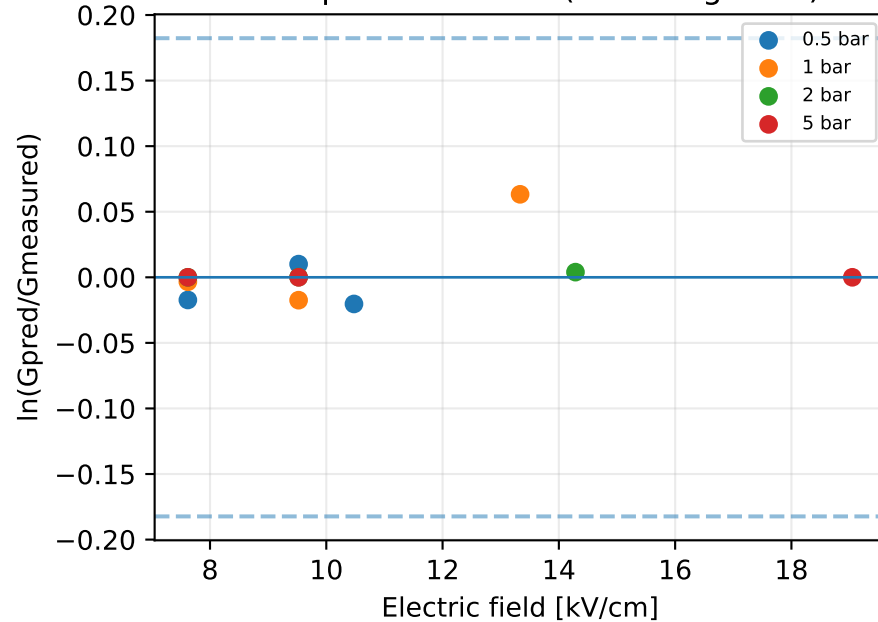
Gain vs electric field



Effective Townsend coefficient vs electric field

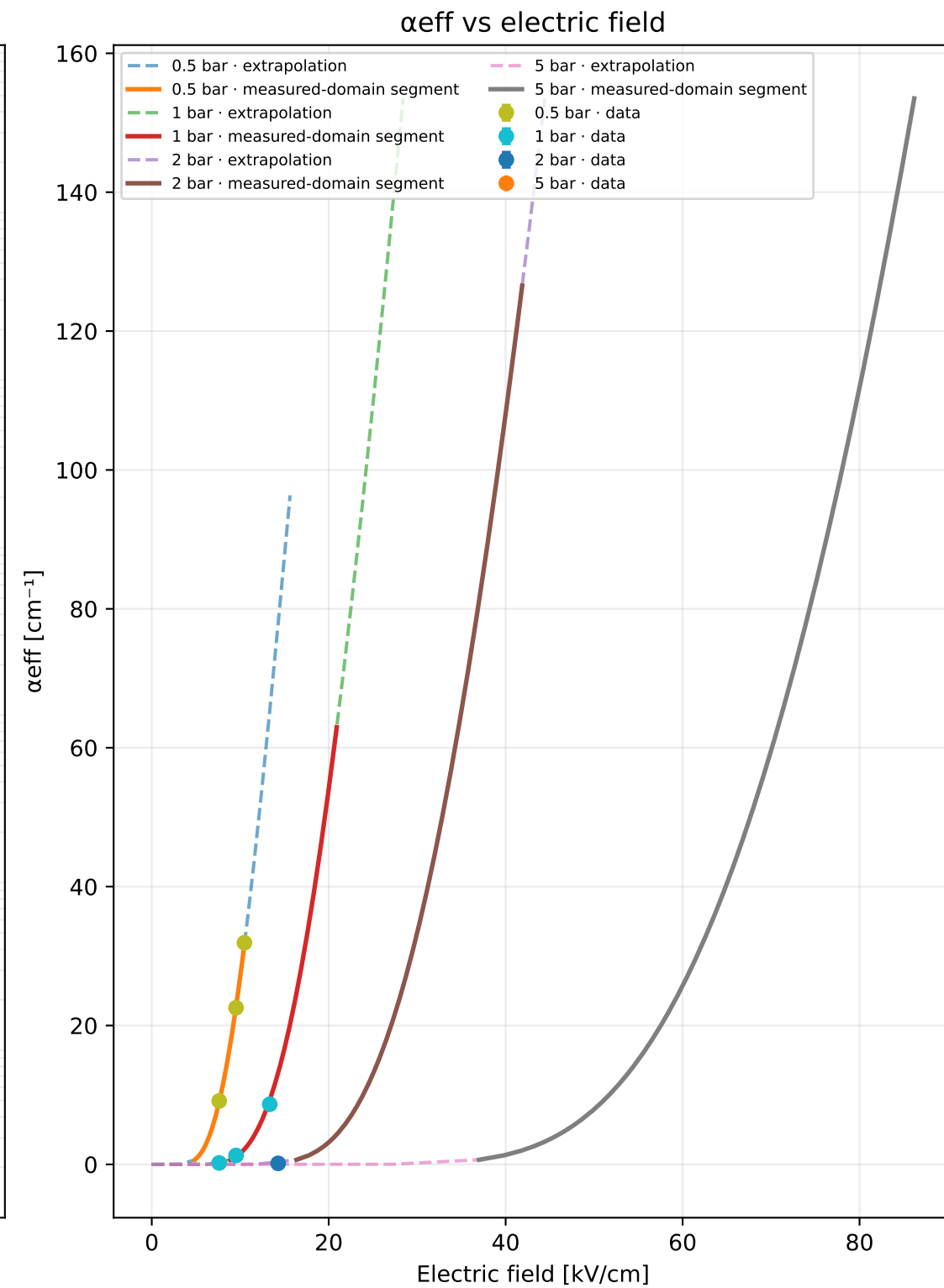
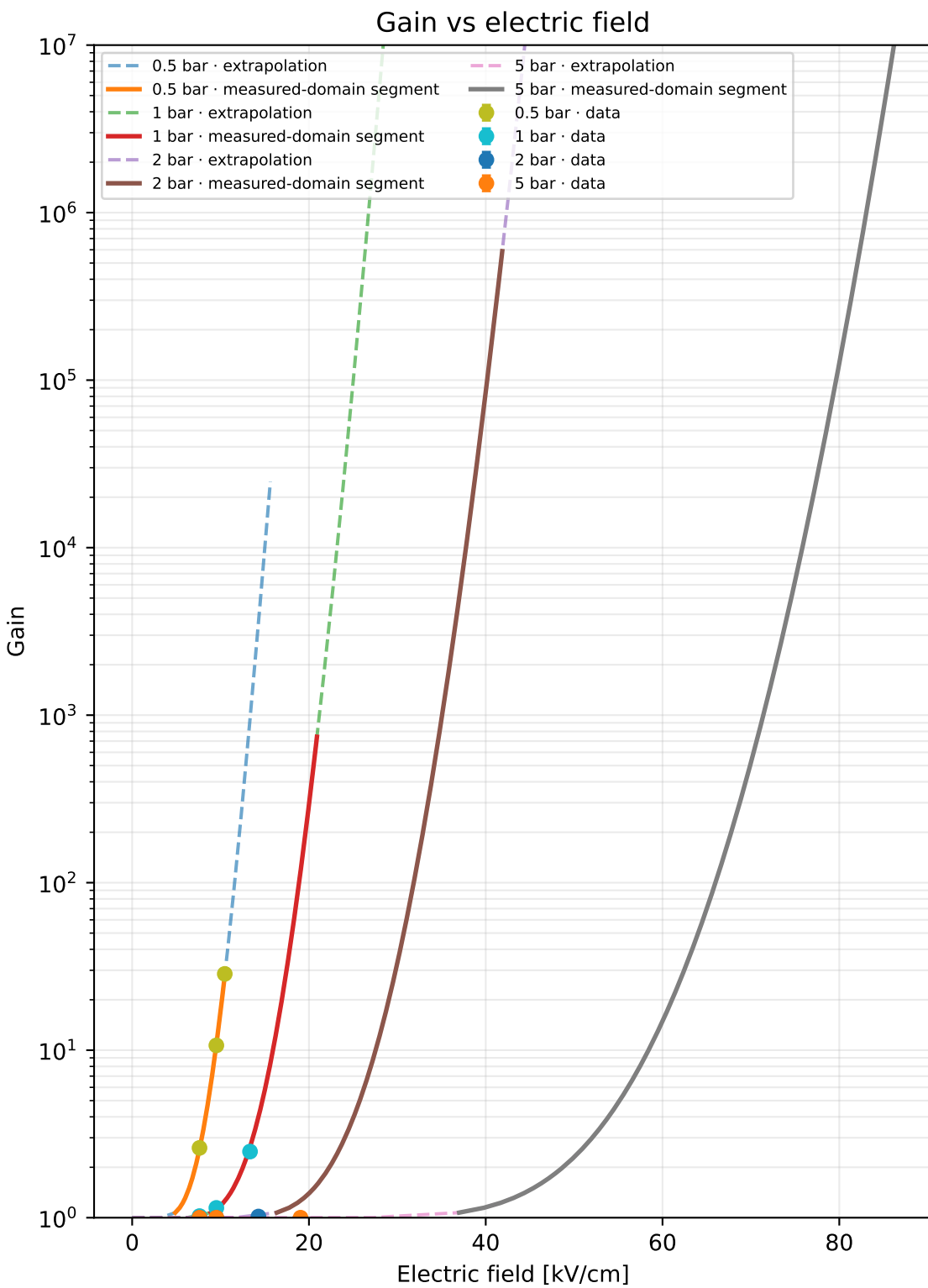


Gain-space residuals ($\pm \ln 1.2$ guides)



ROOT runs: 12
 Unique physical points: 12
 Pressures: 0.5, 1, 2, 5 bar
 Selected model: Townsend 2p
 Status: accepted
 Training RMSE: $\times 1.128$
 Cross-validation median: $\times 1.026$
 Cross-validation worst: $\times 1.105$
 Reduced-field range: 7.143–20.95 kV cm^{-1} bar $^{-1}$
 A=1840.6, B=70.561, m=0, n=1
 Covariance condition: 543
 0.5 bar display limit: G=2.48e+04 (local inversion limit)
 1 bar display limit: G=1e+07 (requested cap)
 2 bar display limit: G=1e+07 (requested cap)
 5 bar display limit: G=1e+07 (requested cap)

Extrapolation capped at $G=1e+07$ or at the local model limit



p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
0.5	7.61905	15.2381	2.61	0.0809	9.13667	500	1
0.5	9.52381	19.0476	10.68	0.455	22.5559	500	1
0.5	10.4762	20.9524	28.534	1.17	31.9152	500	1
1	7.61905	7.61905	1.022	0.00657	0.207252	500	1
1	9.52381	9.52381	1.144	0.0174	1.28125	500	1
1	13.3333	13.3333	2.482	0.0814	8.65776	500	1
2	7.61905	3.80952	1	0	—	500	1
2	9.52381	4.7619	1	0	—	500	1
2	14.2857	7.14286	1.016	0.00629	0.151175	500	1
5	7.61905	1.52381	1	0	—	500	1
5	9.52381	1.90476	1	0	—	500	1
5	19.0476	3.80952	1	0	—	500	1

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-22.6	×1.03	×1.1	yes	—
Townsend 3p	3	-13.1	—	—	no	poor cross validation, large cross validation outlier